

Classification & Optimisation

ACTL3143 & ACTL5111 Deep Learning for Actuaries

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Overview

In these slides, we'll start by giving some demonstrations of training classification models that: 1) predict a binary outcome, then 2) predict a categorical outcome with > 2 options or levels.

Next, we'll step into the maths of how these classification models make predictions, then go look at the high-level ideas of how to “train” them, then finally look at the maths of this training process.

Imports needed for these demos

```
1 import random
2 from pathlib import Path
3
4 import matplotlib.pyplot as plt
5 import numpy as np
6 import pandas as pd
7 import seaborn as sns
8
9 from keras.models import Sequential
10 from keras.layers import Dense, Input
11 from keras.callbacks import EarlyStopping
12
13 from sklearn.datasets import load_iris
14 from sklearn.model_selection import train_test_split
15 from sklearn.preprocessing import OneHotEncoder, StandardScaler
16 from sklearn.compose import make_column_transformer
17 from sklearn.impute import SimpleImputer
18 from sklearn.metrics import confusion_matrix, RocCurveDisplay, PrecisionRecallDisplay
19 from sklearn import set_config
20
21 set_config(transform_output="pandas")
```

Lecture Outline

- **Binary Classification**
- Multiclass Classification
- Dense Layers in Matrices
- Optimisation
- Loss and Derivatives

Stroke Prediction Data description

1. **id**: unique identifier
2. **gender**: “Male”, “Female” or “Other”
3. **age**: age of the patient
4. **hypertension**: 0 or 1 if the patient has hypertension
5. **heart_disease**: 0 or 1 if the patient has any heart disease
6. **ever_married**: “No” or “Yes”
7. **work_type**: “children”, “Govt_jov”, “Never_worked”, “Private” or “Self-employed”
8. **Residence_type**: “Rural” or “Urban”
9. **avg_glucose_level**: average glucose level in blood
10. **bmi**: body mass index
11. **smoking_status**: “formerly smoked”, “never smoked”, “smokes” or “Unknown”
12. **stroke**: 0 or 1 if the patient had a stroke

Source: Kaggle, **Stroke Prediction Dataset**.

Load up the (pre-)preprocessed data

```

1 PROCESSED_DATA_DIR = Path("stroke/processed")
2
3 X_train = pd.read_csv(PROCESSED_DATA_DIR / "x_train.csv")
4 X_val= pd.read_csv(PROCESSED_DATA_DIR / "x_val.csv")
5 X_test = pd.read_csv(PROCESSED_DATA_DIR / "x_test.csv")
6 y_train = pd.read_csv(PROCESSED_DATA_DIR / "y_train.csv")
7 y_val = pd.read_csv(PROCESSED_DATA_DIR / "y_val.csv")
8 y_test = pd.read_csv(PROCESSED_DATA_DIR / "y_test.csv")
9
10 X_train

```

	gender_Female	gender_Male	ever_married_No	ever_married_Yes	Residence_type_Rural	Residence_type_Urban
0	0.0	1.0	0.0	1.0	1.0	0.0
1	0.0	1.0	1.0	0.0	1.0	0.0
2	0.0	1.0	1.0	0.0	1.0	0.0
...	...	...	...	...	...	...
3063	1.0	0.0	0.0	1.0	1.0	0.0
3064	1.0	0.0	0.0	1.0	1.0	0.0
3065	0.0	1.0	1.0	0.0	0.0	1.0

3066 rows × 20 columns

Target variable

```
1 y_train
```

stroke	
0	0
1	0
2	0
...	...
3063	0
3064	0
3065	0

3066 rows × 1 columns

```
1 classes, counts = np.unique(y_train.values.ravel(), return_counts=True)
2 print("Classes:", classes)
3 print("Counts:", counts)
```

Classes: [0 1]
Counts: [2909 157]

Setup a binary classification model

```

1 def create_model(seed=42):
2     random.seed(seed)
3     model = Sequential()
4     model.add(Input(X_train.shape[1:]))
5     model.add(Dense(32, "leaky_relu"))
6     model.add(Dense(16, "leaky_relu"))
7     model.add(Dense(1, "sigmoid"))
8     return model

```

```

1 model = create_model()
2 model.summary()

```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 32)	672
dense_1 (Dense)	(None, 16)	528
dense_2 (Dense)	(None, 1)	17

Total params: 1,217 (4.75 KB)
 Trainable params: 1,217 (4.75 KB)
 Non-trainable params: 0 (0.00 B)

Fit the model

```
1 model = create_model()  
2 model.compile("adam", "binary_crossentropy")  
3 model.fit(X_train, y_train, epochs=5, verbose=2)
```

Epoch 1/5

96/96 - 0s - 1ms/step - loss: 0.2734

Epoch 2/5

96/96 - 0s - 716us/step - loss: 0.1753

Epoch 3/5

96/96 - 0s - 715us/step - loss: 0.1665

Epoch 4/5

96/96 - 0s - 717us/step - loss: 0.1619

Epoch 5/5

96/96 - 0s - 737us/step - loss: 0.1595

<keras.src.callbacks.history.History at 0x123577cb0>

Track accuracy as the model trains

```
1 model = create_model()  
2 model.compile("adam", "binary_crossentropy", metrics=["accuracy"])  
3 model.fit(X_train, y_train, epochs=5, verbose=2)
```

Epoch 1/5

96/96 - 0s - 797us/step - accuracy: 0.9204 - loss: 0.2711

Epoch 2/5

96/96 - 0s - 776us/step - accuracy: 0.9488 - loss: 0.1766

Epoch 3/5

96/96 - 0s - 782us/step - accuracy: 0.9488 - loss: 0.1667

Epoch 4/5

96/96 - 0s - 783us/step - accuracy: 0.9488 - loss: 0.1623

Epoch 5/5

96/96 - 0s - 762us/step - accuracy: 0.9488 - loss: 0.1595

<keras.src.callbacks.history.History at 0x1237bce10>

Run a long fit

```
1 model = create_model()  
2 model.compile("adam", "binary_crossentropy", metrics=["accuracy"])  
3 %time hist = model.fit(X_train, y_train, epochs=500, validation_data=(X_val, y_val), verb
```

CPU times: user 45 s, sys: 3.83 s, total: 48.9 s

Wall time: 46.1 s

Add early stopping

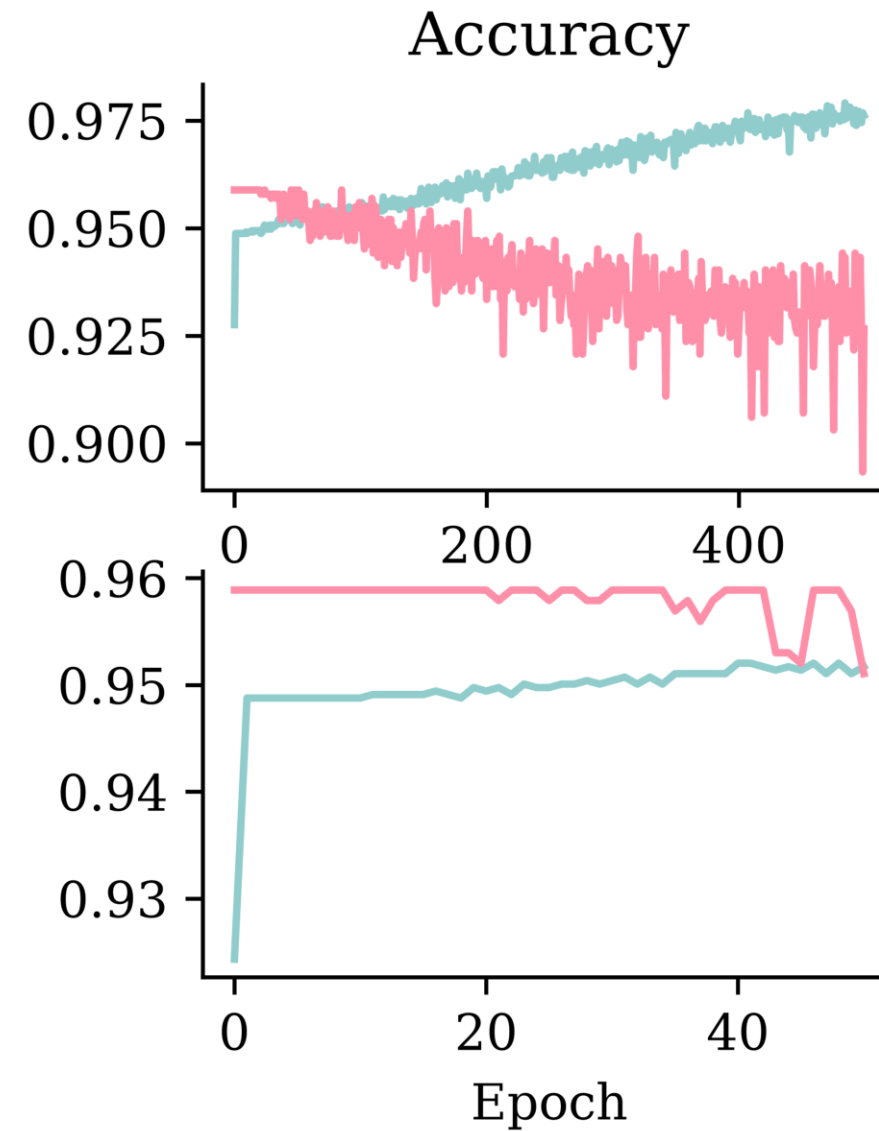
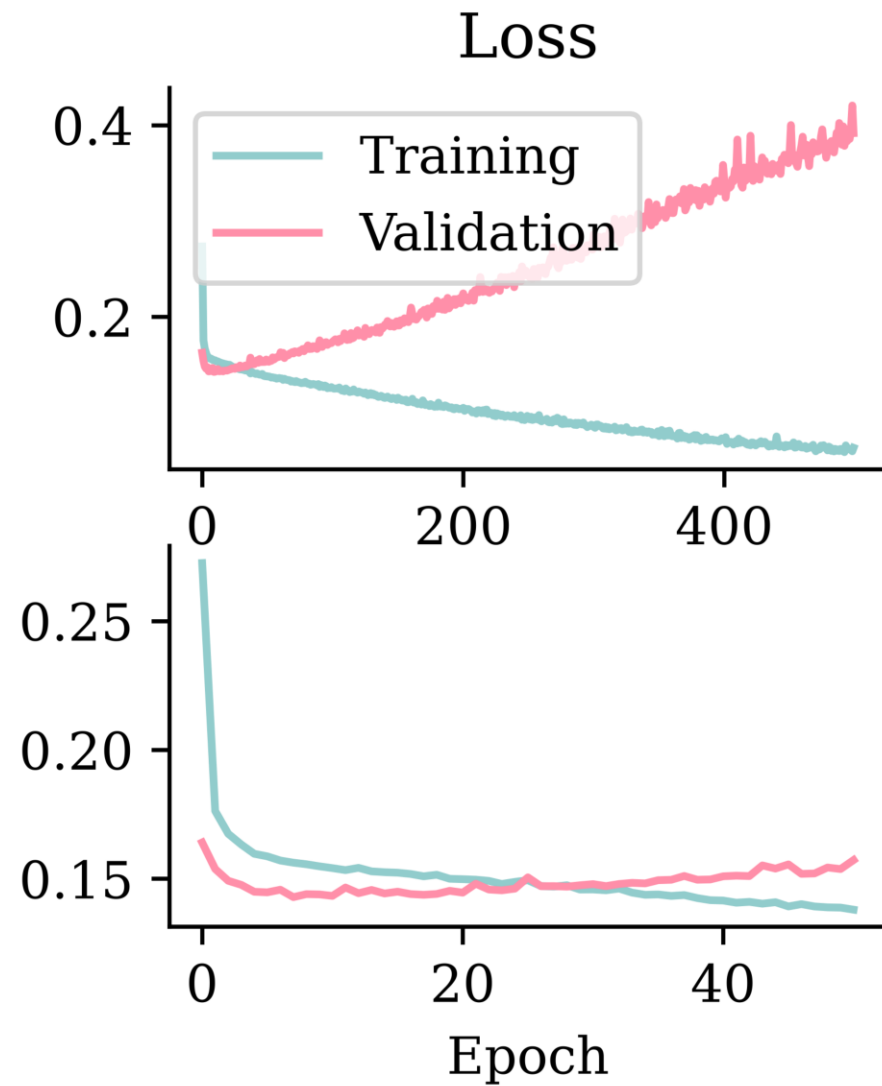
```
1 model = create_model()  
2 model.compile("adam", "binary_crossentropy", metrics=["accuracy"])  
3 es = EarlyStopping(restore_best_weights=True, patience=50, monitor="val_accuracy")  
4 %time hist_es = model.fit(X_train, y_train, epochs=500, validation_data=(X_val, y_val), c  
5 print(f"Stopped after {len(hist_es.history['loss'])} epochs.")
```

CPU times: user 4.56 s, sys: 394 ms, total: 4.96 s

Wall time: 4.68 s

Stopped after 51 epochs.

Fitting metrics



Add metrics, compile, and fit

```

1 model = create_model()
2
3 pr_auc = keras.metrics.AUC(curve="PR", name="pr_auc")
4 model.compile(optimizer="adam", loss="binary_crossentropy",
5               metrics=[pr_auc, "accuracy", "auc"])
6
7 es = EarlyStopping(patience=50, restore_best_weights=True,
8                   monitor="val_pr_auc", verbose=1)
9 model.fit(X_train, y_train, callbacks=[es], epochs=1_000, verbose=0,
10          validation_data=(X_val, y_val));

```

Epoch 81: early stopping

Restoring model weights from the end of the best epoch: 31.

```

1 model.evaluate(X_val, y_val, verbose=0)

```

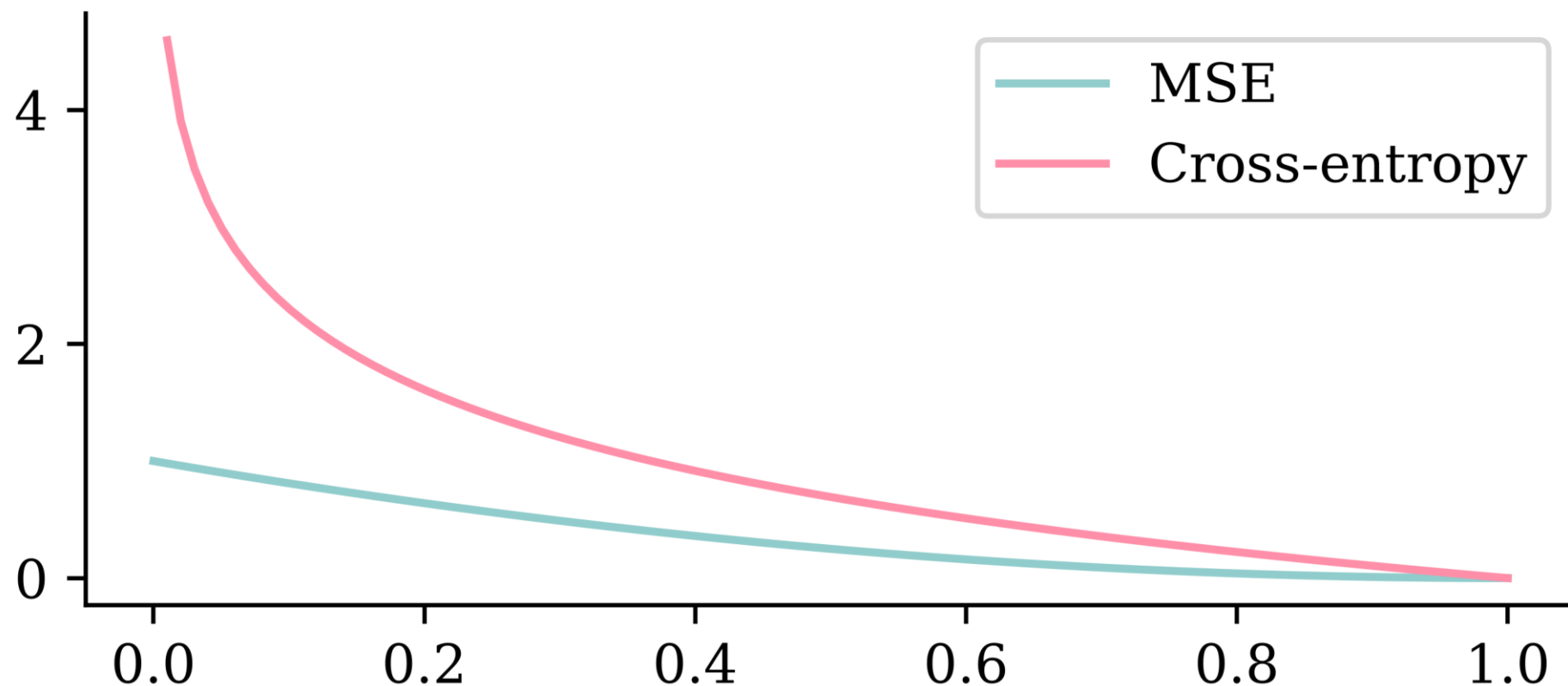
```

[0.14898666739463806,
 0.12857568264007568,
 0.9569471478462219,
 0.8119411468505859]

```

Why use cross-entropy loss?

```
1 p = np.linspace(0, 1, 100)
2 plt.plot(p, (1 - p) ** 2)
3 plt.plot(p, -np.log(p))
4 plt.legend(["MSE", "Cross-entropy"]);
```



Overweight the minority class

```

1 model = create_model()
2
3 pr_auc = keras.metrics.AUC(curve="PR", name="pr_auc")
4 model.compile(optimizer="adam", loss="binary_crossentropy",
5               metrics=[pr_auc, "accuracy", "auc"])
6
7 es = EarlyStopping(patience=50, restore_best_weights=True,
8                   monitor="val_pr_auc", verbose=1)
9 model.fit(X_train, y_train.to_numpy(), callbacks=[es], epochs=1_000, verbose=0,
10          validation_data=(X_val, y_val), class_weight={0: 1, 1: 10});

```

Epoch 64: early stopping

Restoring model weights from the end of the best epoch: 14.

```
1 model.evaluate(X_val, y_val, verbose=0)
```

```
[0.3523019552230835,
0.13380154967308044,
0.7896282076835632,
0.8259596824645996]
```

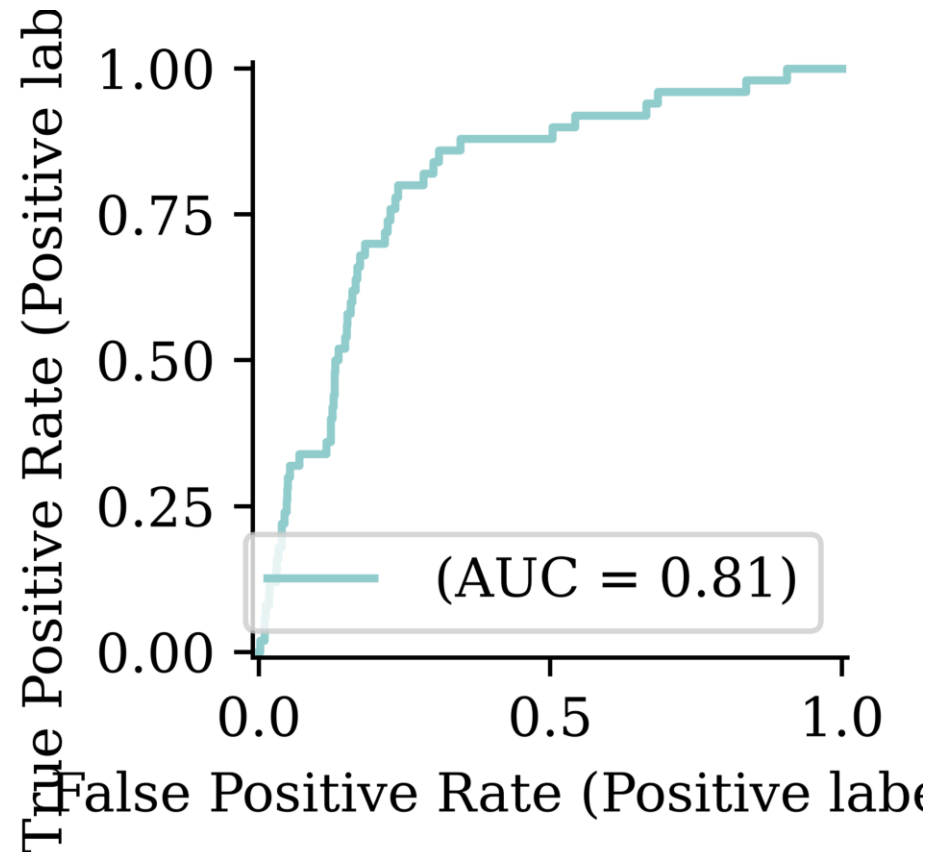
```
1 model.evaluate(X_test, y_test, verbose=0)
```

```
[0.36996063590049744,
0.15842117369174957,
0.7954990267753601,
0.8060390949249268]
```

Classification Metrics

```
1 y_pred = model.predict(X_test, verbose=0)
```

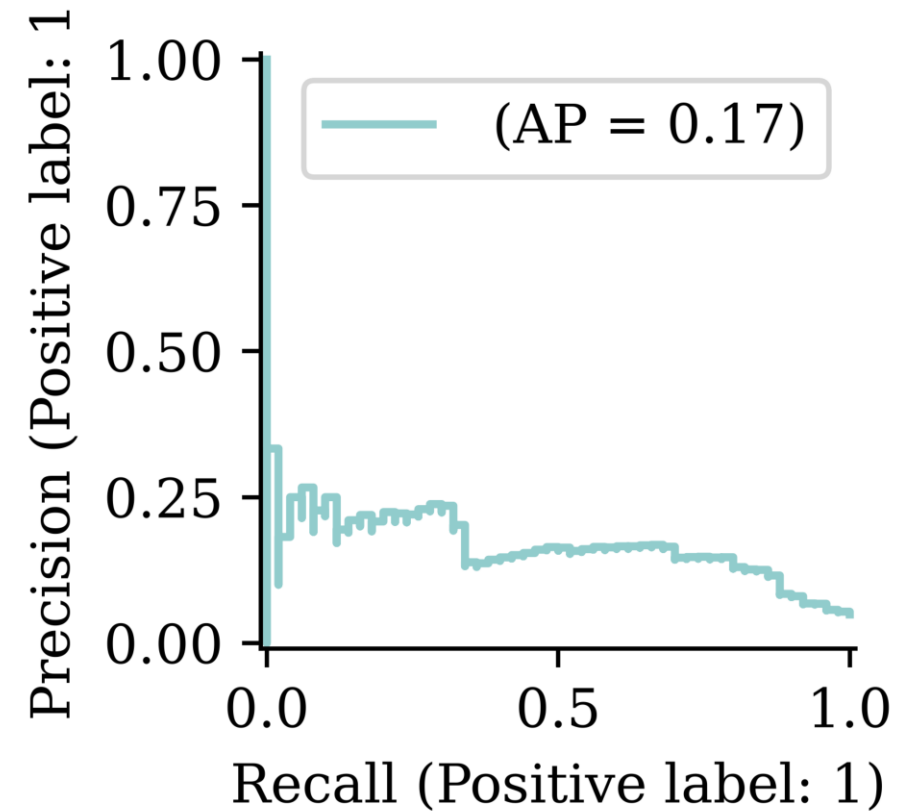
```
1 RocCurveDisplay.from_predictions(y_test, y_pred, name="");
```



```
1 y_pred_stroke = y_pred > 0.5
2 confusion_matrix(y_test, y_pred_stroke)
```

```
array([[778, 194],
       [ 15,  35]])
```

```
1 PrecisionRecallDisplay.from_predictions(y_test, y_pred, name="");
```



```
1 y_pred_stroke = y_pred > 0.3
2 confusion_matrix(y_test, y_pred_stroke)
```

```
array([[647, 325],
       [  7,  43]])
```

Lecture Outline

- Binary Classification
- **Multiclass Classification**
- Dense Layers in Matrices
- Optimisation
- Loss and Derivatives

Iris dataset

```

1 iris = load_iris()
2 names = ["SepalLength", "SepalWidth", "PetalLength", "PetalWidth"]
3 features = pd.DataFrame(iris.data, columns=names)
4 features

```

	SepalLength	SepalWidth	PetalLength	PetalWidth
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
...	...	...	...	...
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

Target variable

```
1 iris.target_names
```

```
array(['setosa', 'versicolor', 'virginica'],  
      dtype='<U10')
```

```
1 iris.target[:8]
```

```
array([0, 0, 0, 0, 0, 0, 0, 0])
```

```
1 target = iris.target  
2 target = target.reshape(-1, 1)  
3 target[:8]
```

```
array([[0],  
       [0],  
       [0],  
       [0],  
       [0],  
       [0],  
       [0]])
```

```
1 classes, counts = np.unique(  
2     target,  
3     return_counts=True  
4 )  
5 print(classes)  
6 print(counts)
```

```
[0 1 2]  
[50 50 50]
```

```
1 iris.target_names[  
2     target[[0, 30, 60]]  
3 ]
```

```
array(['setosa',  
       'setosa',  
       'versicolor'], dtype='<U10')
```

Split the data into train and test

```
1 X_train, X_test, y_train, y_test = train_test_split(features, target, random_state=24)
2 X_train
```

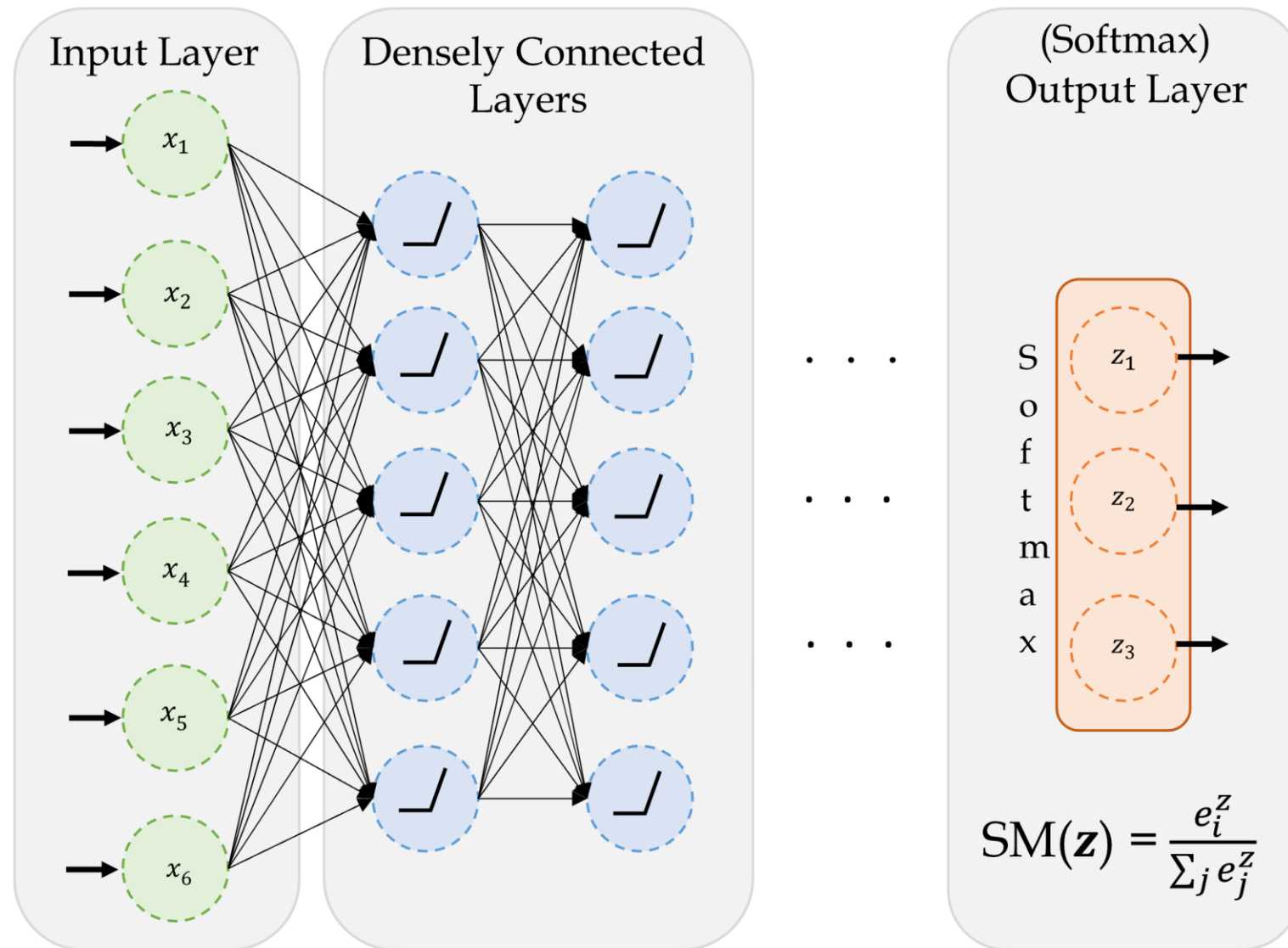
	SepalLength	SepalWidth	PetalLength	PetalWidth
53	5.5	2.3	4.0	1.3
58	6.6	2.9	4.6	1.3
95	5.7	3.0	4.2	1.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
87	6.3	2.3	4.4	1.3
131	7.9	3.8	6.4	2.0

112 rows $\times$ 4 columns

```
1 X_test.shape, y_test.shape
```

((38, 4), (38, 1))

A basic classifier network



A basic network for classifying into three categories.

Source: Marcus Lautier (2022).

Create a classifier model

```
1 NUM_FEATURES = len(features.columns)
2 NUM_CATS = len(np.unique(target))
3
4 print("Number of features:", NUM_FEATURES)
5 print("Number of categories:", NUM_CATS)
```

Number of features: 4

Number of categories: 3

Make a function to return a Keras model:

```
1 def build_model(seed=42):
2     random.seed(seed)
3     return Sequential([
4         Dense(30, activation="relu"),
5         Dense(NUM_CATS, activation="softmax")
6     ])
```

Fit the model

```
1 model = build_model()  
2 model.compile("adam", "sparse_categorical_crossentropy")  
3  
4 model.fit(X_train, y_train, epochs=5, verbose=2);
```

Epoch 1/5

4/4 - 0s - 1ms/step - loss: 1.3920

Epoch 2/5

4/4 - 0s - 947us/step - loss: 1.2912

Epoch 3/5

4/4 - 0s - 965us/step - loss: 1.2196

Epoch 4/5

4/4 - 0s - 985us/step - loss: 1.1576

Epoch 5/5

4/4 - 0s - 886us/step - loss: 1.1084

Track accuracy as the model trains

```
1 model = build_model()  
2 model.compile("adam", "sparse_categorical_crossentropy", metrics=["accuracy"])  
3 model.fit(X_train, y_train, epochs=5, verbose=2);
```

Epoch 1/5

4/4 - 0s - 1ms/step - accuracy: 0.2857 - loss: 1.3930

Epoch 2/5

4/4 - 0s - 970us/step - accuracy: 0.2857 - loss: 1.2970

Epoch 3/5

4/4 - 0s - 978us/step - accuracy: 0.2857 - loss: 1.2203

Epoch 4/5

4/4 - 0s - 986us/step - accuracy: 0.2946 - loss: 1.1596

Epoch 5/5

4/4 - 0s - 935us/step - accuracy: 0.3393 - loss: 1.1067

Run a long fit

```
1 model = build_model()  
2 model.compile("adam", "sparse_categorical_crossentropy", \  
3               metrics=["accuracy"])  
4 %time hist = model.fit(X_train, y_train, epochs=500, \  
5               validation_split=0.25, verbose=False)
```

CPU times: user 1.43 s, sys: 300 ms, total: 1.73 s

Wall time: 1.55 s

Evaluation now returns both *loss* and *accuracy*.

```
1 model.evaluate(X_test, y_test, verbose=False)
```

```
[0.08639740198850632, 0.9736841917037964]
```

Add early stopping

```

1 model_es = build_model()
2 model_es.compile("adam", "sparse_categorical_crossentropy", \
3                 metrics=["accuracy"])
4
5 es = EarlyStopping(restore_best_weights=True, patience=50,
6                   monitor="val_accuracy")
7 %time hist_es = model_es.fit(X_train, y_train, epochs=500, \
8                             validation_split=0.25, callbacks=[es], verbose=False);
9
10 print(f"Stopped after {len(hist_es.history['loss'])} epochs.")

```

CPU times: user 205 ms, sys: 44 ms, total: 249 ms

Wall time: 222 ms

Stopped after 70 epochs.

Evaluation on test set:

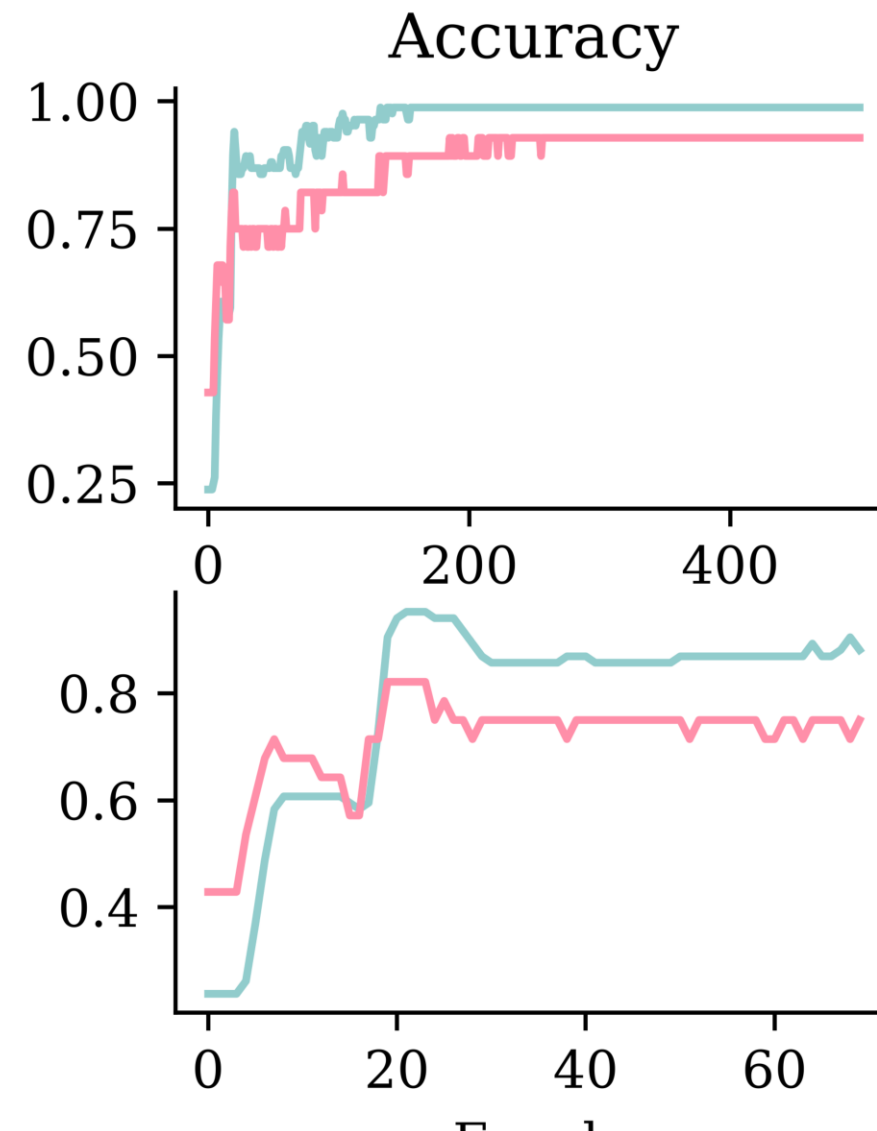
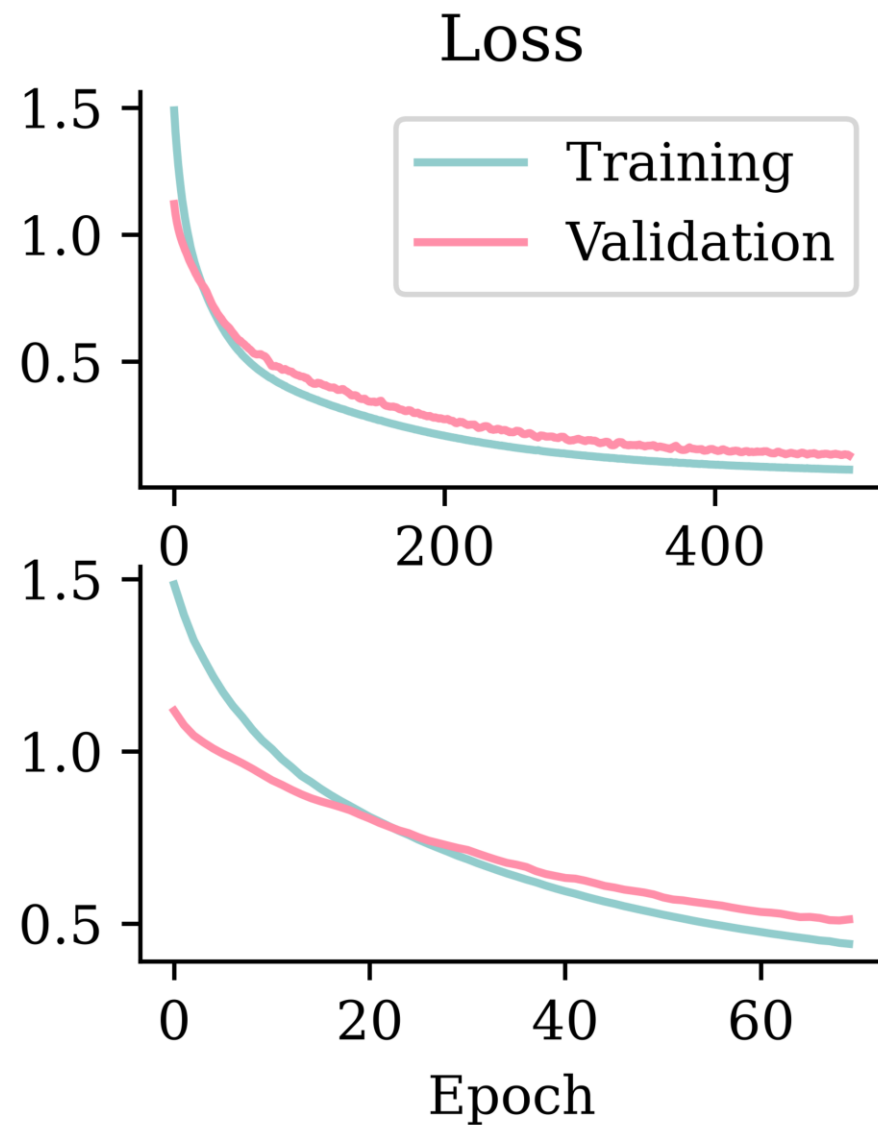
```

1 model_es.evaluate(X_test, y_test, verbose=False)

```

[0.8077937960624695, 0.9210526347160339]

Fitting metrics



What is the softmax activation?

It creates a “probability” vector: $\text{Softmax}(\mathbf{x})_i = \frac{e^{x_i}}{\sum_j e^{x_j}}$.

In NumPy:

```
1 out = np.array([5, -1, 6])  
2 (np.exp(out) / np.exp(out).sum()).round(3)
```

```
array([0.27, 0.  , 0.73])
```

In Keras:

```
1 out = keras.ops.convert_to_tensor([[5.0, -1.0, 6.0]])  
2 keras.ops.round(keras.ops.softmax(out), 3)
```

```
tensor([[0.2690, 0.0010, 0.7310]])
```

Prediction using classifiers

```
1 y_test[:4]
```

```
array([[2],
       [2],
       [1],
       [1]])
```

```
1 y_pred = model.predict(X_test.head(4), verbose=0)
2 y_pred
```

```
array([[2.02e-06, 7.64e-02, 9.24e-01],
       [1.86e-07, 1.62e-03, 9.98e-01],
       [1.44e-02, 9.76e-01, 1.00e-02],
       [2.80e-03, 8.50e-01, 1.48e-01]], dtype=float32)
```

```
1 # Add 'keepdims=True' to get a column vector.
2 np.argmax(y_pred, axis=1)
```

```
array([2, 2, 1, 1])
```

```
1 iris.target_names[np.argmax(y_pred, axis=1)]
```

```
array(['virginica', 'virginica', 'versicolor', 'versicolor'], dtype='<U10')
```

Summary: Classification models in Keras

If the number of classes is c , then:

Target	Output Layer	Loss Function
Binary ($c = 2$)	1 neuron with <code>sigmoid</code> activation	Binary Cross-Entropy
Multi-class ($c > 2$)	c neurons with <code>softmax</code> activation	Categorical Cross-Entropy

Summary: Optionally output logits

If the number of classes is c , then:

Target	Output Layer	Loss Function
Binary ($c = 2$)	1 neuron with <code>linear</code> activation	Binary Cross-Entropy (<code>from_logits=True</code>)
Multi-class ($c > 2$)	c neurons with <code>linear</code> activation	Categorical Cross-Entropy (<code>from_logits=True</code>)

Summary: Code examples

Binary

```
1 model = Sequential([
2     # Skipping the earlier layers
3     Dense(1, activation="sigmoid")
4 ])
5 model.compile(loss="binary_crossentropy")
```

Binary (logits)

```
1 model = Sequential([
2     # Skipping the earlier layers
3     Dense(1, activation="linear")
4 ])
5 loss = BinaryCrossentropy(from_logits=True)
6 model.compile(loss=loss)
```

Multi-class

```
1 model = Sequential([
2     # Skipping the earlier layers
3     Dense(n_classes, activation="softmax")
4 ])
5 model.compile(loss="sparse_categorical_crossentropy")
```

Multi-class (logits)

```
1 model = Sequential([
2     # Skipping the earlier layers
3     Dense(n_classes, activation="linear")
4 ])
5 loss = SparseCategoricalCrossentropy(from_logits=True)
6 model.compile(loss=loss)
```

Both `BinaryCrossentropy` and `SparseCategoricalCrossentropy` live in `keras.losses`.

Lecture Outline

- Binary Classification
- Multiclass Classification
- **Dense Layers in Matrices**
- Optimisation
- Loss and Derivatives

Logistic regression

Observations: $\mathbf{x}_{i,\bullet} \in \mathbb{R}^2$.

Target: $y_i \in \{0, 1\}$.

Predict: $\hat{y}_i = \mathbb{P}(Y_i = 1)$.

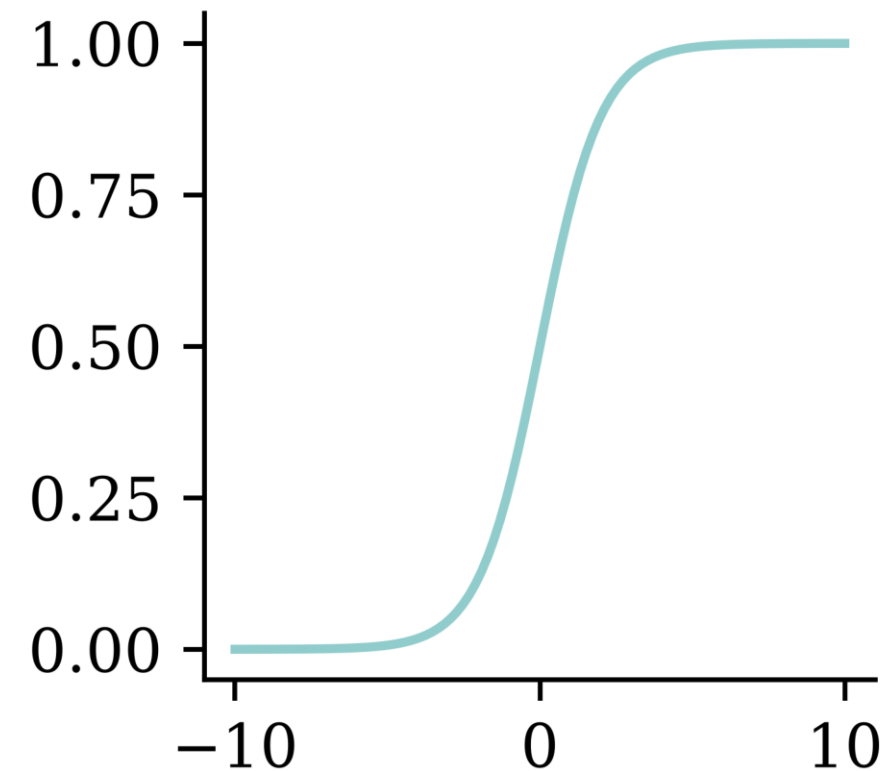
The model

For $\mathbf{x}_{i,\bullet} = (x_{i,1}, x_{i,2})$:

$$z_i = x_{i,1}w_1 + x_{i,2}w_2 + b$$

$$\hat{y}_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}.$$

```
1 x = np.linspace(-10, 10, 100)
2 y = 1/(1 + np.exp(-x))
3 plt.plot(x, y);
```



Multiple observations

```
1 data = pd.DataFrame({"x_1": [1, 3, 5], "x_2": [2, 4, 6], "y": [0, 1, 1]})
2 data
```

	x_1	x_2	y
0	1	2	0
1	3	4	1
2	5	6	1

Let $w_1 = 1$, $w_2 = 2$ and $b = -10$.

```
1 w_1 = 1; w_2 = 2; b = -10
2 data["x_1"] * w_1 + data["x_2"] * w_2 + b
```

```
0    -5
1     1
2     7
dtype: int64
```

Matrix notation

Have $\mathbf{X} \in \mathbb{R}^{3 \times 2}$.

```
1 X_df = data[["x_1", "x_2"]]
2 X = X_df.to_numpy()
3 X
```

```
array([[1, 2],
       [3, 4],
       [5, 6]])
```

Let $\mathbf{w} = (w_1, w_2)^\top \in \mathbb{R}^{2 \times 1}$.

```
1 w = np.array([[1], [2]])
2 w
```

```
array([[1],
       [2]])
```

$$\mathbf{z} = \mathbf{X}\mathbf{w} + b, \quad \mathbf{a} = \sigma(\mathbf{z})$$

```
1 z = X.dot(w) + b
2 z
```

```
array([[ -5],
       [  1],
       [  7]])
```

```
1 1 / (1 + np.exp(-z))
```

```
array([[0.01],
       [0.73],
       [1.   ]])
```

Using a softmax output

Observations: $\mathbf{x}_{i,\bullet} \in \mathbb{R}^2$. Predict: $\hat{y}_{i,j} = \mathbb{P}(Y_i = j)$.

Target: $\mathbf{y}_{i,\bullet} \in \{(1, 0), (0, 1)\}$.

The model: For $\mathbf{x}_{i,\bullet} = (x_{i,1}, x_{i,2})$

$$z_{i,1} = x_{i,1}w_{1,1} + x_{i,2}w_{2,1} + b_1,$$

$$z_{i,2} = x_{i,1}w_{1,2} + x_{i,2}w_{2,2} + b_2.$$

$$\hat{y}_{i,1} = \text{Softmax}_1(\mathbf{z}_i) = \frac{e^{z_{i,1}}}{e^{z_{i,1}} + e^{z_{i,2}}},$$

$$\hat{y}_{i,2} = \text{Softmax}_2(\mathbf{z}_i) = \frac{e^{z_{i,2}}}{e^{z_{i,1}} + e^{z_{i,2}}}.$$

Multiple observations

1 data

	x_1	x_2	y_1	y_2
0	1	2	1	0
1	3	4	0	1
2	5	6	0	1

Choose:

$$w_{1,1} = 1, w_{2,1} = 2,$$

$$w_{1,2} = 3, w_{2,2} = 4, \text{ and}$$

$$b_1 = -10, b_2 = -20.$$

```
1 w_11 = 1; w_21 = 2; b_1 = -10
2 w_12 = 3; w_22 = 4; b_2 = -20
3 data["x_1"] * w_11 + data["x_2"] * w_21 + b_1
```

0 -5

1 1

2 7

dtype: int64

Matrix notation

Have $\mathbf{X} \in \mathbb{R}^{3 \times 2}$.

```
1 X
```

```
array([[1, 2],
       [3, 4],
       [5, 6]])
```

$\mathbf{W} \in \mathbb{R}^{2 \times 2}$, $\mathbf{b} \in \mathbb{R}^2$

```
1 W = np.array([[1, 3], [2, 4]])
2 b = np.array([-10, -20])
3 display(W); b
```

```
array([[1, 3],
       [2, 4]])
```

```
array([-10, -20])
```

$$\mathbf{Z} = \mathbf{XW} + \mathbf{b}, \quad \mathbf{A} = \text{Softmax}(\mathbf{Z}).$$

```
1 Z = X @ W + b
2 Z
```

```
array([[-5, -9],
       [ 1,  5],
       [ 7, 19]])
```

```
1 np.exp(Z) / np.sum(np.exp(Z),
2   axis=1, keepdims=True)
```

```
array([[9.82e-01, 1.80e-02],
       [1.80e-02, 9.82e-01],
       [6.14e-06, 1.00e+00]])
```

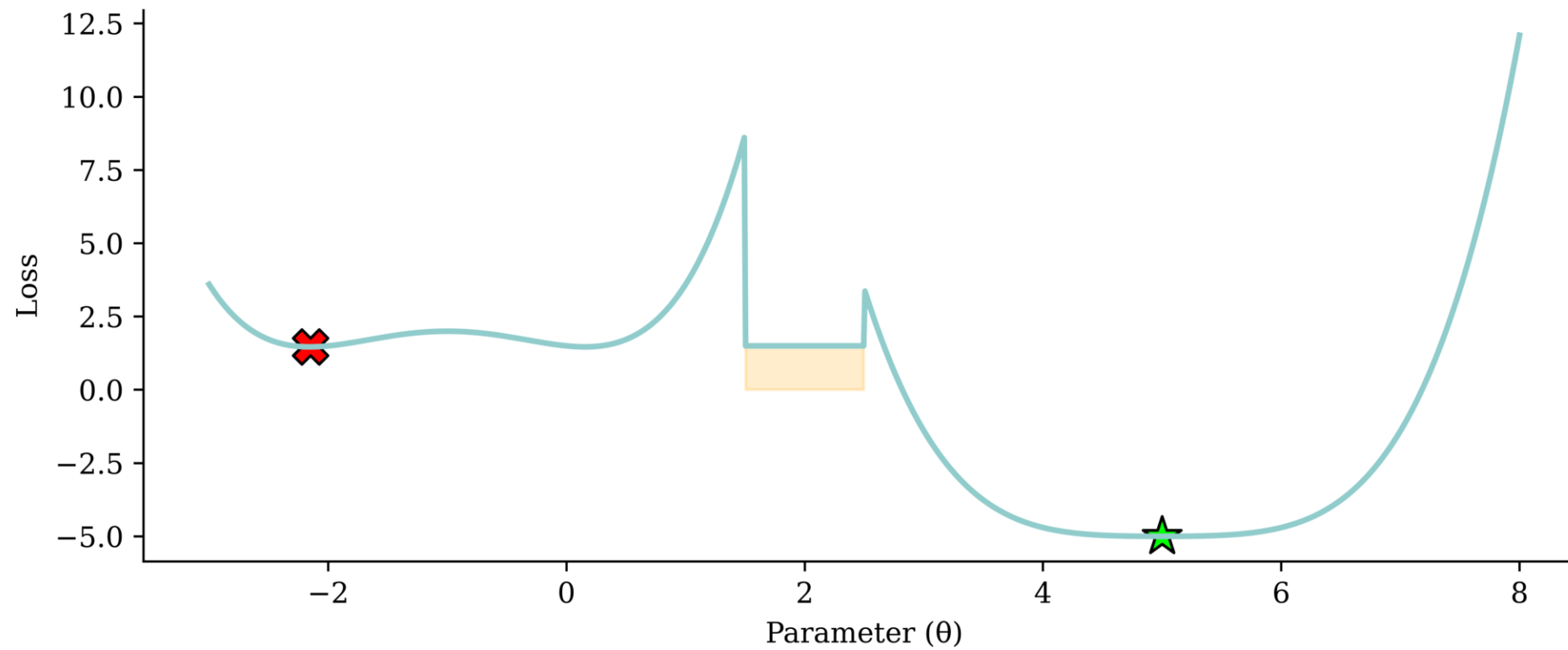
Lecture Outline

- Binary Classification
- Multiclass Classification
- Dense Layers in Matrices
- **Optimisation**
- Loss and Derivatives

Gradient-based learning

In-class demo

Gradient descent pitfalls



★ Global minimum $\times$ Local minimum Plateau

Go over all the training data

Called *batch gradient descent*.

```
1 for i in range(num_epochs):  
2     gradient = evaluate_gradient(loss_function, data, weights)  
3     weights = weights - learning_rate * gradient
```

Pick a random training example

Called *stochastic gradient descent*.

```
1 for i in range(num_epochs):  
2     rnd.shuffle(data)  
3     for example in data:  
4         gradient = evaluate_gradient(loss_function, example, weights)  
5         weights = weights - learning_rate * gradient
```

Take a group of training examples

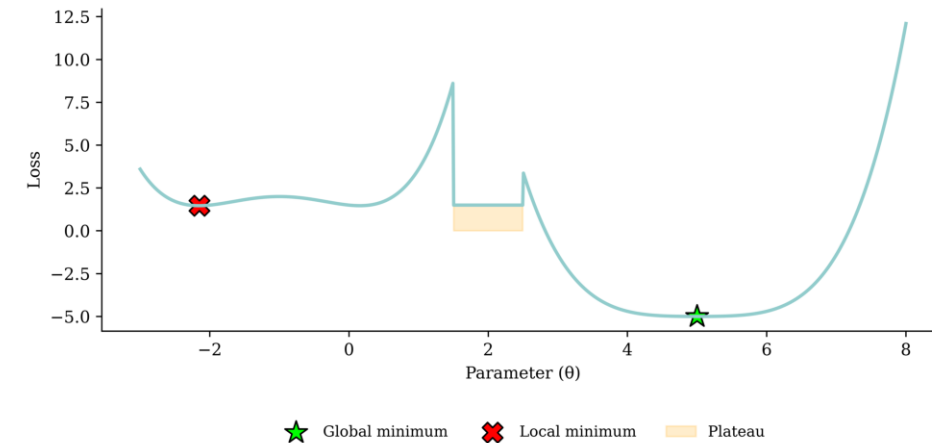
Called *mini-batch gradient descent*.

```
1 for i in range(num_epochs):  
2     rnd.shuffle(data)  
3     for b in range(num_batches):  
4         batch = data[b * batch_size : (b + 1) * batch_size]  
5         gradient = evaluate_gradient(loss_function, batch, weights)  
6         weights = weights - learning_rate * gradient
```

Mini-batch gradient descent

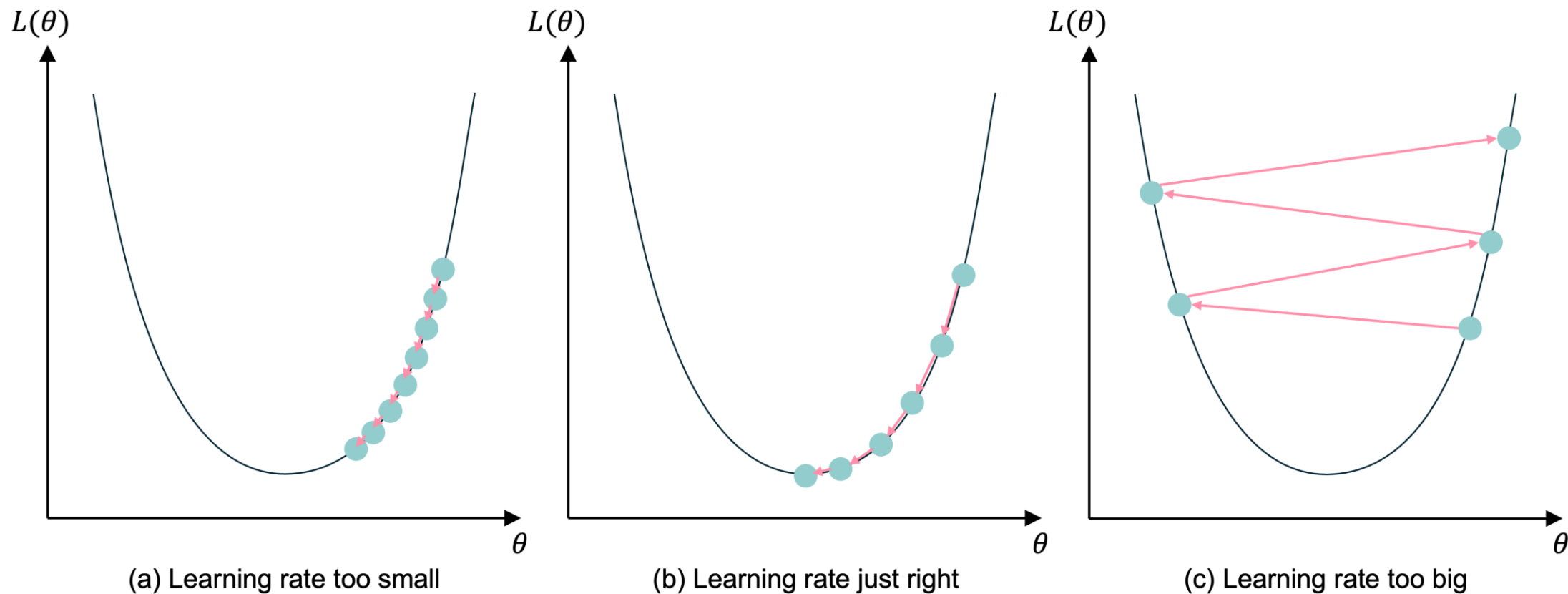
Why?

1. Because we have to (data is too big to shove it all in a single batch)
2. Because it is faster (lots of quick noisy steps *takes longer* than a few slow super accurate steps)
3. The noise helps us jump out of local minima



Noisy gradient means we might jump out of a local minimum.

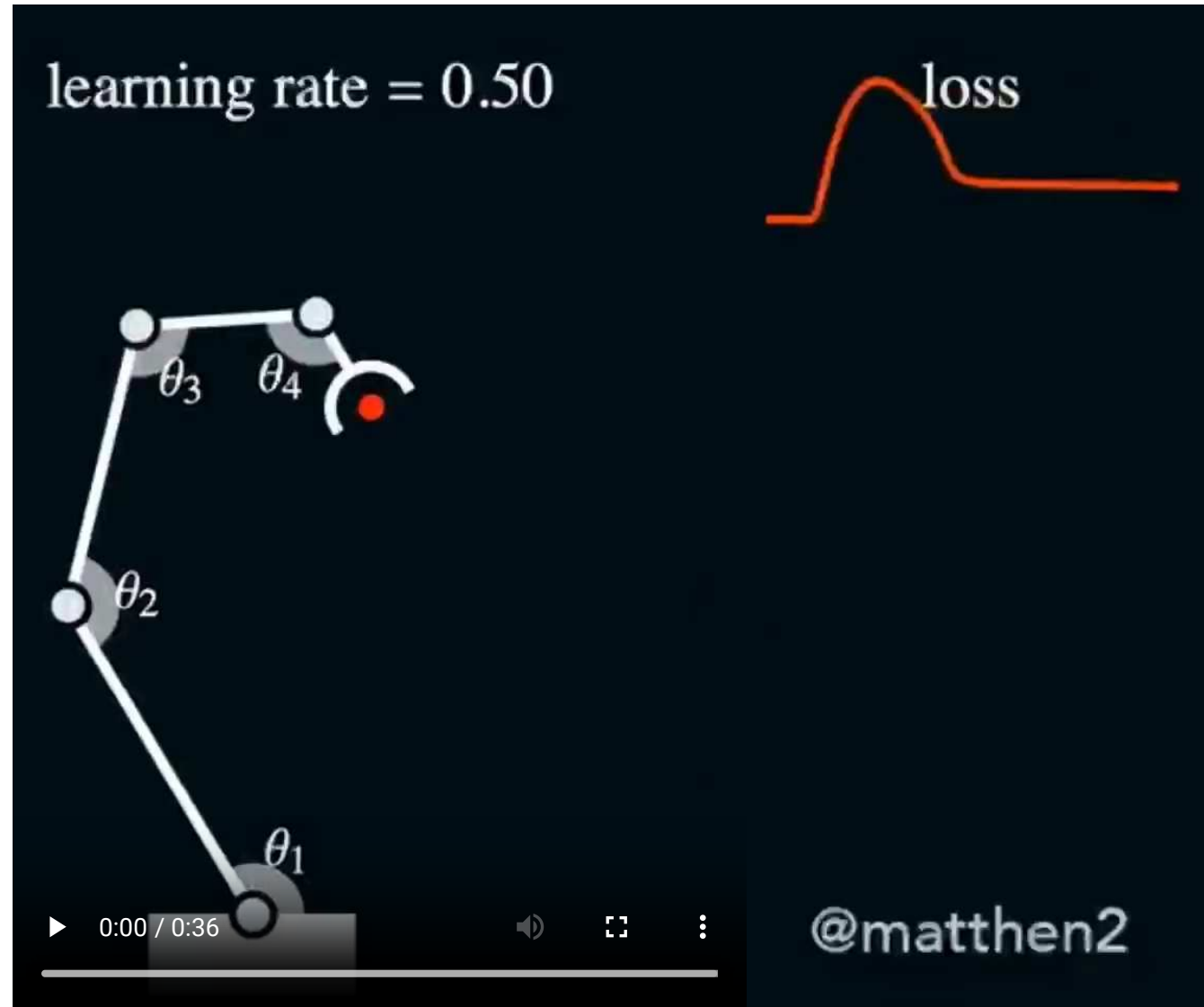
Learning rates



Gradient descent with different learning rates

Source: Melissa Renard (2025), adapted from Jeremy Jordan (2018), *Setting the learning rate of your neural network*.

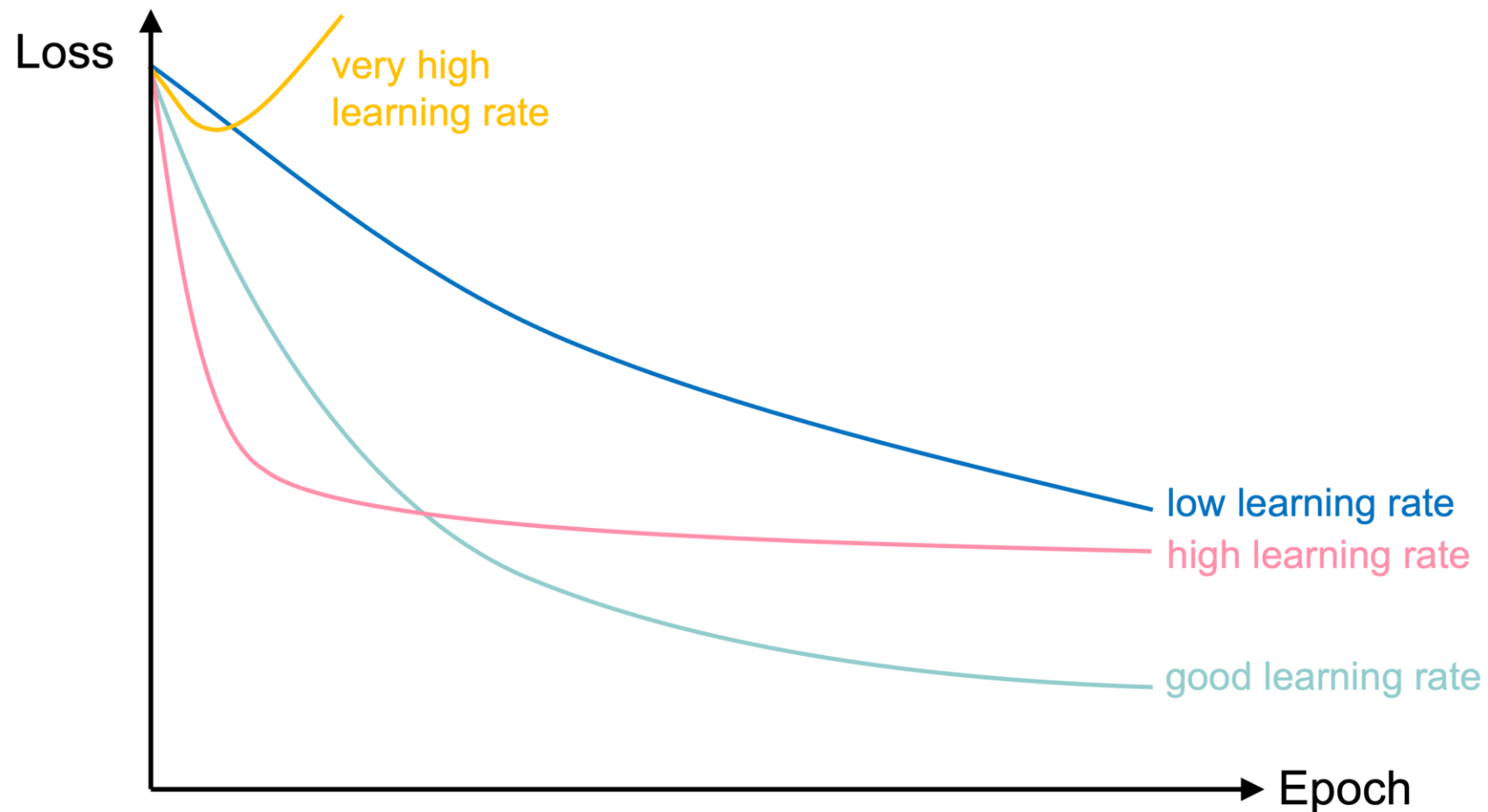
Learning rates #2



Changing the learning rates for a robot arm.

Source: Matt Henderson (2021), [X post](#)

Learning rate schedule



Learning curves for various learning rates

In training the learning rate may be tweaked manually.

Source: Melissa Renard (2025), adapted from [Deep Learning for Computer Vision - Stanford Spring 2025](#)

Lecture Outline

- Binary Classification
- Multiclass Classification
- Dense Layers in Matrices
- Optimisation
- **Loss and Derivatives**

Example: linear regression

$$\hat{y}(x) = wx + b$$

For some observation $\{x_i, y_i\}$, the squared error loss is

$$\text{Loss}_i = (\hat{y}(x_i) - y_i)^2$$

For a batch of the first n observations the MSE loss is

$$\text{Loss}_{1:n} = \frac{1}{n} \sum_{i=1}^n (\hat{y}(x_i) - y_i)^2$$

Derivatives

Since $\hat{y}(x) = wx + b$,

$$\frac{\partial \hat{y}(x)}{\partial w} = x \text{ and } \frac{\partial \hat{y}(x)}{\partial b} = 1.$$

As $\text{Loss}_i = (\hat{y}(x_i) - y_i)^2$, we know

$$\frac{\partial \text{Loss}_i}{\partial \hat{y}(x_i)} = 2(\hat{y}(x_i) - y_i).$$

Chain rule

$$\frac{\partial \text{Loss}_i}{\partial \hat{y}(x_i)} = 2(\hat{y}(x_i) - y_i), \quad \frac{\partial \hat{y}(x)}{\partial w} = x, \quad \text{and} \quad \frac{\partial \hat{y}(x)}{\partial b} = 1.$$

Putting this together, we have

$$\frac{\partial \text{Loss}_i}{\partial w} = \frac{\partial \text{Loss}_i}{\partial \hat{y}(x_i)} \times \frac{\partial \hat{y}(x_i)}{\partial w} = 2(\hat{y}(x_i) - y_i) x_i$$

and

$$\frac{\partial \text{Loss}_i}{\partial b} = \frac{\partial \text{Loss}_i}{\partial \hat{y}(x_i)} \times \frac{\partial \hat{y}(x_i)}{\partial b} = 2(\hat{y}(x_i) - y_i).$$

Applying the chain rule *backwards* through the network to get every gradient is *backpropagation* (Rumelhart et al., 1986).

We need non-zero derivatives

This is why can't use accuracy as the loss function for classification.

Also why we can have the *dead ReLU* problem.

Stochastic gradient descent (SGD)

Start with $\theta_0 = (w, b)^\top = (0, 0)^\top$.

Randomly pick $i = 5$, say $x_i = 5$ and $y_i = 5$.

$$\hat{y}(x_i) = 0 \times 5 + 0 = 0 \Rightarrow \text{Loss}_i = (0 - 5)^2 = 25.$$

The partial derivatives are

$$\begin{aligned} \frac{\partial \text{Loss}_i}{\partial w} &= 2(\hat{y}(x_i) - y_i) x_i = 2 \cdot (0 - 5) \cdot 5 = -50, \text{ and} \\ \frac{\partial \text{Loss}_i}{\partial b} &= 2(0 - 5) = -10. \end{aligned}$$

The gradient is $\nabla \text{Loss}_i = (-50, -10)^\top$.

SGD, first iteration

Start with $\boldsymbol{\theta}_0 = (w, b)^\top = (0, 0)^\top$.

Randomly pick $i = 5$, say $x_i = 5$ and $y_i = 5$.

The gradient is $\nabla \text{Loss}_i = (-50, -10)^\top$.

Use learning rate $\eta = 0.01$ to update

$$\begin{aligned}\boldsymbol{\theta}_1 &= \boldsymbol{\theta}_0 - \eta \nabla \text{Loss}_i \\ &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 0.01 \begin{pmatrix} -50 \\ -10 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0.5 \\ 0.1 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.1 \end{pmatrix}.\end{aligned}$$

SGD, second iteration

Start with $\boldsymbol{\theta}_1 = (w, b)^\top = (0.5, 0.1)^\top$.

Randomly pick $i = 9$, say $x_i = 9$ and $y_i = 17$.

The gradient is $\nabla \text{Loss}_i = (-223.2, -24.8)^\top$.

Use learning rate $\eta = 0.01$ to update

$$\begin{aligned}\boldsymbol{\theta}_2 &= \boldsymbol{\theta}_1 - \eta \nabla \text{Loss}_i \\ &= \begin{pmatrix} 0.5 \\ 0.1 \end{pmatrix} - 0.01 \begin{pmatrix} -223.2 \\ -24.8 \end{pmatrix} \\ &= \begin{pmatrix} 0.5 \\ 0.1 \end{pmatrix} + \begin{pmatrix} 2.232 \\ 0.248 \end{pmatrix} = \begin{pmatrix} 2.732 \\ 0.348 \end{pmatrix}.\end{aligned}$$

Batch gradient descent (BGD)

For the first n observations $\text{Loss}_{1:n} = \frac{1}{n} \sum_{i=1}^n \text{Loss}_i$ so

$$\begin{aligned} \frac{\partial \text{Loss}_{1:n}}{\partial w} &= \frac{1}{n} \sum_{i=1}^n \frac{\partial \text{Loss}_i}{\partial w} = \frac{1}{n} \sum_{i=1}^n \frac{\partial \text{Loss}_i}{\hat{y}(x_i)} \frac{\partial \hat{y}(x_i)}{\partial w} \\ &= \frac{1}{n} \sum_{i=1}^n 2(\hat{y}(x_i) - y_i) x_i. \end{aligned}$$

$$\begin{aligned} \frac{\partial \text{Loss}_{1:n}}{\partial b} &= \frac{1}{n} \sum_{i=1}^n \frac{\partial \text{Loss}_i}{\partial b} = \frac{1}{n} \sum_{i=1}^n \frac{\partial \text{Loss}_i}{\hat{y}(x_i)} \frac{\partial \hat{y}(x_i)}{\partial b} \\ &= \frac{1}{n} \sum_{i=1}^n 2(\hat{y}(x_i) - y_i). \end{aligned}$$

BGD, first iteration ($\theta_0 = \mathbf{0}$)

	x	y	y_hat	loss	dL/dw	dL/db
0	1	0.99	0	0.98	-1.98	-1.98
1	2	3.00	0	9.02	-12.02	-6.01
2	3	5.01	0	25.15	-30.09	-10.03

So $\nabla \text{Loss}_{1:3}$ is

```
1 nabra = np.array([df["dL/dw"].mean(), df["dL/db"].mean()])
2 nabra
```

array([-14.69, -6.])

so with $\eta = 0.1$ then θ_1 becomes

```
1 theta_1 = theta_0 - 0.1 * nabra
2 theta_1
```

array([1.47, 0.6])

BGD, second iteration

	x	y	y_hat	loss	dL/dw	dL/db
0	1	0.99	2.07	1.17	2.16	2.16
1	2	3.00	3.54	0.29	2.14	1.07
2	3	5.01	5.01	0.00	-0.04	-0.01

So $\nabla \text{Loss}_{1:3}$ is

```
1 nabra = np.array([df["dL/dw"].mean(), df["dL/db"].mean()])
2 nabra
```

```
array([1.42, 1.07])
```

so with $\eta = 0.1$ then θ_2 becomes

```
1 theta_2 = theta_1 - 0.1 * nabra
2 theta_2
```

```
array([1.33, 0.49])
```

Package Versions

```
1 from watermark import watermark
2 print(watermark(python=True, packages="keras,matplotlib,numpy,pandas,seaborn,scipy,torch")
```

Python implementation: CPython

Python version : 3.14.5

IPython version : 9.15.0

keras : 3.15.0

matplotlib: 3.11.0

numpy : 2.5.0

pandas : 3.0.3

seaborn : 0.13.2

scipy : 1.18.0

torch : 2.12.1

Recommended viewing

Some very easy-to-follow explanations of these topics, plus catchy tunes:

- Softmax and argmax
- Cross entropy
- Cross entropy derivatives and backpropagation

Glossary

- accuracy
- classification problem
- confusion matrix
- cross-entropy loss
- metrics
- sigmoid activation function
- softmax activation
- batch gradient descent
- batches, batch size
- global minimum, local minimum
- gradient-based learning, hill-climbing
- learning rate, learning rate schedule
- plateau
- stochastic gradient descent
- mini-batch gradient descent

References

Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323(6088), 533–536.